



Gus: Gas Planner

Quote

I see the important role that natural gas will play globally and, more importantly, the important role it will play in the U.S. in terms of meeting future energy demand.

Motivation

- Provide upcoming plans with little regulatory scrutiny
- Ensure proper gas storage levels through the heating season
- Least cost gas is provided to members and customers

Trusts information from

- His predecessor's models
- His spreadsheets that he uses to collect and validate data
- Some accounting systems
- Contracts

Sphere of Influence

Accounting, Gas Dispatch

Brands and Influencers

- Accounting has a pretty strong influence on Gus and many times Gus has to adjust his models to support decisions made by Accounting
- Gus' alma mater
- His friends and family

Likes

- Dig into challenging problems
- MS Excel
- Not being on call outside normal business hours
- Spending time with his young family
- Home improvement projects

Dislikes

- Having accounting dictate the best plans to "justify" with his models.
- Internal I.T. roadblocks
- Long running models that require updating over the weekend
- "Pixel Hunts" due to bad data inputs.
- Business Travel

Study Horizon

Long Term, Mid Term

EE Product knowledge

Medium

Education

Masters Degree in Industrial Engineering

Experience

5 years

Description

Gus works at a North American gas utility that provides natural gas heating to homes, business, and industrial plants. He is responsible for the utilities annual PGA filing ensuring that gas customers are getting low cost heating delivered reliably. He is also responsible for evaluating supply, transportation, and storage contracts each year along with expansion studies done once every 5 years.

His workflow outlined below:

- Gather input data necessary to conduct the next round of studies including
 - All current contract data
 - Gas storage inventories
 - Load forecast projections or weather projections
 - Price forecast
- He must then get agreement on the load forecasts used for modelling including buy in from all key stakeholders. These forecasts can be based on 10, 20, or 30 year historical lows for reliability studies.
- Execute models that reflect different scenarios,
 - colder than normal winter
 - expected cold winter
 - average winter
 - expected warm winter
 - colder than normal winter
 - Total HDD studies vs Peak day studies
- Aggregate results and create PGA report
- File
- Respond with any questions from regulator

Gus sees opportunities to work with power planners to negotiate more competitive contracts with suppliers. There are internal roadblocks that he is unsure how to navigate.

Goals

- Provide PGA filings / Rate Cases with limited responses from regulatory body
- Find creative ways to save the company more money and lower heating costs to their customers
- Ensure there is enough gas to serve his customers
- Take over all department planning responsibilities including the dispatch center

Challenges

- Input data assumptions
- Accounting department making decisions before his analysis is done
- Over-constrained system
- Having proper hardware resources available when necessary

Responsibilities

- Provide PGA (purchase gas adjustment) filings to the local regulatory body every year
- Validate assumptions to regulatory body
- Ensure properly sized mid and long term supply, transportation, and storage contracts
- Ensure gas storage targets are hit every spring
- Ensure reliable gas is provided at least cost to the customer base
- Look after long term gas price forecast models and short term load forecasting

Skills

- Strong analytical skills
- Good problem solver
- Some technical skills including the ability to write complex Excel Macros
- Has a both a broad and deep understanding of his companies business

User Type

4) Experienced user who needs to have product knowledge to justify results

Trends Impacting Persona

- Gas shale has provided Gus new and different options to serve gas.
- Reduction in coal power production has put more pressure on gas demand (reduced net carbon output)
- Excess supply and increased demand puts pressure on deliverability of gas to Gus's citygate

User Personality

- Detailed minded
- Enjoys complex data and modelling problems
- Professional
- Polite
- Diligent



Quote
Luck is great, but most of life is hard work.

Motivation
Ambition

Company growth and ownership in the company

The opportunity to work on many different tasks

Trusts information from
Anyone that has data

Sphere of Influence
Company owners

Brands and Influencers
Anyone that can make his job easier

Likes
Uniqueness of job responsibilities

That he is reporting to the CTO of the company

He has equity in the company and he is on the ground floor of a small company

Playing golf on Sunday and spending time with his friends

Dislikes
When the company misses payroll

The long hours

Getting chewed out by the owner of the company

Poorly outlined responsibilities

Study Horizon
Mid Term, Short Term, Long Term

EE Product knowledge
Low

Education
Bachelors Degree in Business Management

Experience
5 years

Description
Mark joined 5 years ago after working odd jobs for a few years out of college. The job is very difficult with many responsibilities; however, Mark is never bored. He is required to manage many data streams, models, internal processes, and at times technical tasks. He rarely has time to do any one thing well, but he is able to get just enough done to move on to the next thing.

The company doesn't have money for tools and data to help him do his job better so he does a lot work to manually get the data and maintain this on his own. He also must learn how to build models and maintain these as well; which he isn't very good at. Some of his outputs from his models have negatively impacted the company's bottom line, but Mark is hard worker and because of this the company tolerates the occasional mistake.

Mark enjoys playing golf when he has the time as well as hanging out with his friends on the weekends.

Goals
Aspirations of the company growing and getting rich

Challenges
Become more competitive in the cost of energy to grow the customer base

Meet ownerships expectations

Maintain all existing systems while working to continually improve with little to no help

Responsibilities
Acquire all the data necessary to model the system served by the marketing company

Create and maintain the models that are used to minimize energy rates to the customer base

Find ways to be more competitive and sign better deals for energy delivery

Convince the company owners to invest in better tools to improve their competitive situation

Skills
Hard worker

Utility infielder -- broad skill set

Committed to success

User Type
2) Needs a limited output set to support his/her responsibilities

Trends Impacting Persona

User Personality



Mary: Market Analyst



Quote
"The devil is in the details - you have to have the right data, modeled correctly, to get valuable insights on trends and expectations"

EE Product knowledge
Medium

Education
Bachelor of Science in Business Administration

Experience
8 years

- Motivation**
- Compensation
 - Recognition as a key contributor
 - Career growth
 - Personal development

Description
Mary is a market analyst for Big Vertical Utility. She was raised in smalltown, Montana and was always the star of her class. She accepted an academic scholarship to West Coast U where she finished with above average marks in Business Administration. Mary's dream was to work for Goldman-Sachs but came to Big Firm when that didn't work out.

Mary has steadily learned the principles of the power markets and is looking to continued development professionally to break the glass ceiling.

Trusts information from
Team members, market experts

Mary's workflow is largely project based:

- Get particulars for new analysis (new asset investment, new contract, etc)
- Determine appropriate footprint for analysis
- Request update ffrom subject matter experts or appropriate data for analysis
- Receive and incorporate new data
- Perform analyses
- Present results internally

Sphere of Influence
Traders, Risk Management, Power Plant Developers

- Goals**
- Have tools and skills to improve forecast accuracy (economic and physical asset performance)
 - Co-optimization of market drivers to improve results
 - Be able to complete forecasts more quickly; cycle time is important
 - User Experience; Easily embed latest market data into model
 - Be recognized for her contributions to the company bottom-line

- Brands and Influencers**
- Traders
 - Plant Developers
 - Other market analysts
 - Saks 5th Avenue
 - Total Wine
 - BMW

- Challenges**
- Must wait for data from other departments
 - Must defend forecast against historical market performance and/or expectation of future market
 - Benchmarking performance
 - Would like to work with a team that understands the market rules and how to model them
 - Not enough historical data to accurately backcast models
 - Validation of model capturing binding constraints in the markets

- Likes**
- Time saving technologies
 - Wine
 - Theater

- Dislikes**
- Wasting time
 - People who act like they know more than they do

- Responsibilities**
- Providing accurate market forecasts for asset/contract performance to facilitate decision making
 - Provides analysis recommending optimal acquisitions/divestitures
 - Provides analyses facilitating fuel decision making
 - Model expected policy/regulation and understand impacts on portfolio
 - Collaborates with fuels, plants, trade floor, finance, load forecasters, demand side, transmission, etc

Study Horizon
Long Term, Mid Term, Short Term

- Skills**
- Identifying new technology that will give her an edge
 - Analyzing performance-related data
 - Economics

User Type
4) Experienced user who needs to have product knowledge to justify results

- Trends Impacting Persona**
- Renewable penetration
 - Energy storage modelling
 - most significantly the impact on fossil viability
 - Policy changes

User Personality
Diligent, methodical, communicates well, impatient for analyses to complete/others to provide updated data



Otto: Operator

Quote
"I keep the lights on - if I can save a few bucks that's great"

EE Product knowledge
Low

Education
High School Grad

Experience
20 years

- Motivation**
- Avoid Surprises
 - Adhere to procedure
 - Have reliable information

Trusts information from
his control panel

Sphere of Influence
Operators, Planners

- Brands and Influencers**
- Church
 - The Weather Channel
 - Leads youth group
 - Sports

- Likes**
- Leave shift on time with system in good shape
 - All processes laid out neatly where only the steps necessary are exposed and decisions require very little analysis.
 - Getting bbq brought in during his shift for all the operators/ Taco Tuesday.
 - Boating and other alone time

- Dislikes**
- Having to go 'out of the playbook' to handle system events
 - Getting bad load forecasts, which disrupt the dispatch and create emergency events.
 - Learning new systems or changes in process.
 - Having an emergency forcing him not to miss his fishing trip

Study Horizon
Short Term

Description
Otto is a central dispatch operator for vertical utility XYZ. He joined the Navy out of high school and was assigned to engine room duty. After putting in his 20 years, he retired from the Navy and joined the company.

He wears jeans and worn out t-shirts to work and typically has to work 3 days a week 12 hour shifts. He still has to man the desk overnight every third shift.

Otto enjoys boating and camping as often as he can and gets a lot of discretionary time with working the operator desk.

Otto's day to day task include

- manning the operations desk
- dispatching generating assets each hour
- regular calls with the plant operators scheduled and unscheduled
- monitoring weather events
- coordinating with the short term trading desk
- working with the analysts to improve process and improve load forecasting

Otto's workflow is repetitive and happens each hour:

- He must get the plant status from each facility
- Ensure that the updated load forecast is reasonable
- Validate all current operating conditions
- Validate the automatically generated dispatch for each of the scenarios
- Publish the agreed upon set points to the control system

- Goals**
- Reliably commit and dispatch system
 - Maintain schedule, interchange requirements
 - Staying out of emergency mode
 - Minimize costs if you can

- Challenges**
- Balancing reliability and cost
 - Scope and accuracy of information
 - System response to controls not behaving as simulated
 - Dealing with uncertainty, such as plant outages and poor forecasts

- Responsibilities**
- Ensures input data is current and correct, relies on a high degree of automation
 - Operate the system with reliability as the first priority and economics as a second
 - This includes commit, dispatch, granting of maintenance, acquisition/sale of power as needed.
 - Performs tasks against an internally defined rigorous process with reviews of performance
 - Has ability to operate within guidelines and, using judgement, override some suggested decisions in complex situations
 - He is the central hub of all plant status across the fleet.

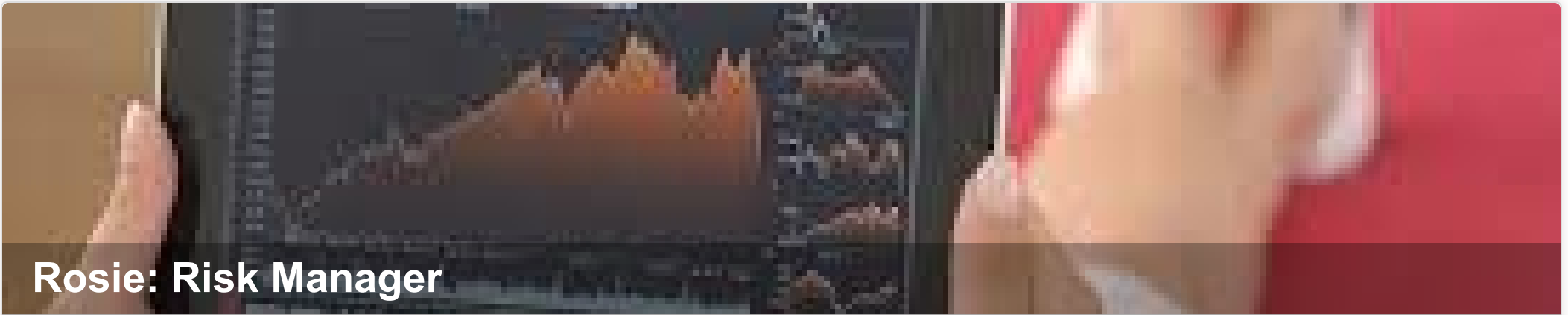
- Skills**
- Following procedure and chain of command
 - Understanding a playbook

User Type
3) Experienced user but doesn't want a lot of product knowledge to support his/her responsibilities

Trends Impacting Persona
More automation in the system gives Otto a sense of uneasiness and distrust of management

Dislikes change due to the rigid nature of his job, especially the high impact of renewables in the system

User Personality
Takes his job very seriously and recognizes the magnitude of doing his job well. Extremely punctual and is never late for anything. Not particularly approachable unless you are blue collar (If you meet him on his boat, at church, or at the gun range is very friendly). A bit crass and not overly ambitious.



Rosie: Risk Manager

Quote

Motivation

Set, monitor and implement,and manage tolerance limits within the corporate policies.

Trusts information from

- Planners
- CFO
- Risk Officer

Sphere of Influence

- Chief Financial Officer
- VP - Planning
- VP - Operations
- VP - Trading
- Risk Officer

Brands and Influencers

- Energy Indices
- Bloomberg
- ICE

Likes

- Well defined corporate policies
- liquidity to enter into contracts without exceptions
- Procurement and planners with understanding of the risk limits

Dislikes

- Liquidity issues in the market leading to higher risk contracts
- Having to take exceptions to policies

Study Horizon

MT to LT

EE Product knowledge
None

Education
Masters in Finance/Economics, MBA ,
or a Law degree

Experience
10 years

Description

Rosie is a Risk Manager who supports the Long Term desk in a utility with Generation, which has to enter into long term contracts on Power and fuel.

Goals

- Approve contracts within the limits.
- translate the policies into tangible limits
- report to management
- Limit the exceptions and manage the targets

Challenges

- Exceptions that will lead to violation of the limits, and not being able to manage the limits
- Exceptions that could lead to legal implications

Responsibilities

Help and manage the traders with contract T&Cs within the limits and extend approvals

Skills

- Knowledge of energy industry
- Energy trading
- finance
- a little bit of governing laws

User Type

2) Needs a limited output set to support his/her responsibilities

Trends Impacting Persona

- market liquidity
- Physical delivery
- Changing Renewable conditions

User Personality

- Organized
- Strict adherence to law
- Good with numbers
- Integrity



Tim: Transmission Planner

Quote
"Transmission Planning is a science - Generation Planning is an art. Transmission planning has few degrees of freedom compared to other disciplines; physics dictates how power flows."

- Motivation**
- Discovering significant benefit through robust investigation
 - Seeing projects through approval from planning through to budget approval
 - Regulatory Filing Deadlines
 - Influences significant capital investments
 - Responsibility to Provide adequate system flexibility to system operators
 - Fear of overlooking issues: Impact of issues can cause wide impacting blackouts

Trusts information from
Team members, superiors, system operators and transmission working groups

Sphere of Influence
Rate Case, Risk Management, Integrated resource plan, Generation interconnection(ISO), Transmission Expansion, Market Policy, Reliability and Grid Resiliency

- Brands and Influencers**
- Industry publications and conferences; IEEE
 - Popular Mechanics, Car and Driver, Consumer Reports
 - Strong Electrical Company brands (SIEMENS-PSSE ; PowerWorld)

- Likes**
- Geographical representation of network topology
 - Capturing the physical characteristics to the necessary extent in the model
 - Confidence that his batch runs will complete successfully
 - Actionable feedback if/when his batch runs fail
 - Minimizing manual data manipulation
 - Visualizing input/output

- Dislikes**
- Need to use multiple tools to perform reliability and cost/benefit analyses
 - Difficulty in easily understanding the cause/effect relationship between elements on the system
 - Interrupting his work process to evaluate new generator locations; Frequent responses to internal changes
 - Worrying about work over the weekend
 - Spending too much time customizing the model
 - Merging data from multiple sources and formats

Study Horizon
Long Term, Mid Term

EE Product knowledge
Medium

Education
Bachelor of Science Electrical Engineering

Experience
10 years

Description
Tim is a transmission planner with Big Utility. He was a member of the debate team in HS and graduated from university with a degree in electrical engineering. Tim started with Big Utility out of college and has worked his way up to Senior Planner due to his ability to understand the steady state responses of the power system.

Tim operates on an annual cycle, updating the plans for the system every year. He is an expert with his current tool suite. The process below describes his workflow:

- Update loadflow models to reflect current system and planned generation/demand
- Perform contingency analyses to determine bottlenecks
- Develop several system enhancements to address each problem area
- Perform cost-benefit analysis for each option to determine ideal
- Prepare report(s)
- Communicate new plans to internal and external stakeholders

But Tim wants to do more. He understands that the current methodology could be improved to include probabilistic results as well as economic benefits, but doesn't know how to do that. Runs a larger number of simulations compared to generation planner.

- Goals**
- Create optimal long-term plans for reliability and deliverability
 - Obtain early buy-in on assumptions to minimize work changes
 - Be able to perform comprehensive analyses to leave nothing to doubt
 - Publish work to advance the industry

- Challenges**
- Must wait for inputs from other departments
 - Must take into consideration changing regulatory environments
 - Choosing the right tools
 - Conservative management resistant to change process or tools

- Responsibilities**
- Ensures futures deliverability of demand requirements
 - Understanding the relationship between elements on the system
 - Develop and maintain processes to reduce error and elapsed time while increasing insight
 - Mentoring junior team members
 - Complete FERC filings

- Skills**
- Power Systems
 - Logical approach to problem solving
 - Economic
 - System-wide thinking/impact analysis
 - Aware of Societal benefits

User Type
3) Experienced user but doesn't want a lot of product knowledge to support his/her responsibilities

- Trends Impacting Persona**
- Decision making shifting from reliability only to including the economic impact of decisions
 - Difficult to get capital investments into rate base:Show significant end user benefit

User Personality
Diligent, Communicates well, respectful of others, patient, emotionally stable, Methodical, conservative (reverts to known solutions), inquisitive